

METFORMIN

- FIRST-LINE**
- HEPATIC FACTORY**: BLOCKADE of HEPATIC GLUCONEOGENESIS / PRIMARY MECHANISM
- NAT. OBE.** (No Natural Obesity)
- EMPTYING B12**: CHECK B12 PERIODICALLY
- XR** (Extended Release)
- REDUCES GI SIDE EFFECTS**
- STOP METFORMIN IN THESE SITUATIONS**: GFR <30, CONTRAST, AKI, SEPSIS

2 SGLT2 INHIBITORS

- SGLT2 drain**: BLOCKS RENAL GLUCOSE REABSORPTION (INSULIN-INDEPENDENT)
- SATIURI-URESIS**: ↓ BP / ↓ INTRAGLOMERULAR PRESSURE
- EUGLYCEMIC DKA** — GLUCOSE MAY BE NORMAL
- NOT FOR T1DM**
- ↓ CV MORTALITY**: ↑ CHF HOSPITALIZATIONS, ↓ CKD PROGRESSION

GLP-1 RECEPTOR AGONISTS

- INCRETIN**: INSULIN ↑ ONLY WHEN GLUCOSE IS HIGH
- WEIGHT LOSS**
- CONTRAINDICATED: MTC / MEN2**
- PANCREATITIS**
- STOMACH BRAKE**: SLOWS GASTRIC EMPTYING

4 DPP-4 INHIBITORS

- DPP-4**: INHIBIT DPP-4 → ↑ ENDOGENOUS GLP-1 / LESS POTENT THAN GLP-1RA
- WEIGHT NEUTRAL**
- NO CV BENEFIT**, **NO RENAL PROTECTION**

5 SGLT2 INHIBITORS

- SEMAGLUTIDE** (GLP-1RA)
- FRONT COUNTER** (HbA1c REDUCTION RANKING): METFORMIN (1.8-2.4%), TZD (0.5-1.4%), SGLT2i / GLP-1RA (0.5-1%), DPP4i / ACARBOS (0.5-0.8%)
- HbA1c REDUCTION RANKING**: METFORMIN = MOST POTENT AMONG ORAL AGENTS (TIED WITH SU)

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27 SGLT2 INHIB

1. Metformin — first line

WHAT YOU SEE

Box 1 'Metformin', 1st line; hepatic glucose output reduction; B12 check; XR reduces GI side effects.

WHAT IT ENCODES

Metformin (a biguanide) is first-line oral therapy. Primary action: reduces hepatic gluconeogenesis (activates AMPK), with secondary improvement in peripheral insulin sensitivity. It is weight-neutral, does NOT cause hypoglycemia (no insulin secretion), lowers A1c ~1–1.5%, and is cardioprotective/cheap. Max dose ~2000–2550 mg/day; use extended-release to reduce GI upset; monitor B12 (long-term use → malabsorption/deficiency). Mnemonic: 'metformin muzzles the liver.'

BOARD TRAP

Wrong answer: 'metformin works mainly by stimulating pancreatic insulin secretion.' Discriminator — that is the sulfonylurea/glinide mechanism; metformin's principal action is DECREASING hepatic glucose output and it does not raise insulin secretion (hence no hypoglycemia and no weight gain).

2. Metformin — stop situations

WHAT YOU SEE

STOP: GFR <30, contrast, sepsis/AKI.

WHAT IT ENCODES

Hold/stop metformin when lactate could accumulate: eGFR <30 (contraindicated; caution/dose-reduce 30–45), iodinated IV CONTRAST (hold at the time of/after contrast if eGFR reduced, restart after renal function confirmed stable ~48 h), and sepsis/hypoperfusion/AKI/decompensated HF — all raise lactic acidosis risk. Also hold for major surgery and significant hepatic impairment/alcohol use. Mnemonic: 'metformin + low perfusion = lactic acidosis.'

BOARD TRAP

Wrong answer: 'stop metformin once eGFR falls below 60 (or 45).' Discriminator — the absolute contraindication cutoff is eGFR <30, not 45 or 60 (it may continue with caution/reduced dose at 30–45). Picking 45/60 as the stop point is the classic numeric distractor.

3. SGLT2 inhibitors

WHAT YOU SEE

Box 2 SGLT2i: blocks renal glucose reabsorption; CV mortality/HF/CKD benefit; euglycemic DKA; not for T1DM.

WHAT IT ENCODES

SGLT2 inhibitors (-flozins: empagliflozin, dapagliflozin, canagliflozin) block glucose reabsorption in the proximal tubule → glucosuria, lowering glucose insulin-independently. Benefits: reduced CV death and HF hospitalization (even in non-diabetics), slowed CKD progression/albuminuria, modest weight loss (~2–3 kg) and BP reduction. Risks: euglycemic DKA, genital mycotic infections/UTIs, volume depletion. Mnemonic: 'pee out the sugar, protect heart + kidney.'

BOARD TRAP

Wrong answer: 'add an SGLT2 inhibitor to a type 1 diabetic for extra glucose control.' Discriminator — SGLT2i are not recommended in T1DM because they substantially increase euglycemic DKA risk; reserve them for T2DM (and HF/CKD indications). Also counsel on genital/urinary infections.

4. GLP-1 receptor agonists

WHAT YOU SEE

Box 3 GLP-1 RA: incretin, increases insulin only when glucose high, weight loss; pancreatitis; gastric emptying slowed.

WHAT IT ENCODES

GLP-1 receptor agonists (semaglutide, liraglutide, dulaglutide) are incretin mimetics: they augment GLUCOSE-DEPENDENT insulin secretion, suppress glucagon, slow gastric emptying, and increase satiety → strong weight loss and proven CV (MACE) benefit. Because insulin release is glucose-dependent, monotherapy does not cause hypoglycemia. Main side effects: nausea/vomiting (titrate slowly), pancreatitis risk. Mnemonic: 'smart insulin boost + appetite brake.'

BOARD TRAP

Wrong answer: 'start a GLP-1 RA in a patient with a family history of medullary thyroid carcinoma / MEN2.' Discriminator — personal or family history of medullary thyroid carcinoma or MEN2 is a contraindication (rodent C-cell tumor signal); also avoid with prior pancreatitis. Choose a different class for these patients.

5. DPP-4 inhibitors

WHAT YOU SEE

Box 4 DPP-4i: increases endogenous GLP-1; weight neutral; no CV benefit; no renal protection; less potent than GLP-1 RA.

WHAT IT ENCODES

DPP-4 inhibitors (-gliptins: sitagliptin, linagliptin, saxagliptin) block dipeptidyl peptidase-4, prolonging endogenous GLP-1/GIP → modest glucose-dependent insulin boost. They are weight-NEUTRAL, well tolerated, cause no hypoglycemia, but are weak A1c-lowerers (~0.5–0.8%) with no CV or renal benefit. Linagliptin needs no renal dose adjustment. Mnemonic: 'gliptins = gentle, weak, weight-neutral incretin helpers.'

BOARD TRAP

Wrong answer: 'combine a DPP-4 inhibitor with a GLP-1 RA for additive incretin effect.' Discriminator — both act on the SAME incretin pathway, so the combination is redundant with no added benefit; never co-prescribe them. Also note saxagliptin/alogliptin increased HF hospitalization (avoid in HF).

6. DPP-4 inhibitors detail

WHAT YOU SEE

Box (top): GLP-1 + DPP-4; less potent than GLP-1 RA, no CV benefit, no renal protection.

WHAT IT ENCODES

DPP-4 inhibitors raise endogenous incretins only to physiologic levels, so they are markedly LESS potent than injectable GLP-1 RAs (which achieve supraphysiologic receptor activation) and lack the GLP-1 RA's CV/renal/weight-loss benefits. They are oral, convenient, and hypoglycemia-free, making them an option when injectables aren't feasible and comorbidity benefit isn't the priority. Mnemonic: 'a gliptin is a GLP-1 RA's weaker oral cousin.'

BOARD TRAP

Wrong answer: 'a DPP-4 inhibitor will give the same weight loss and CV protection as a GLP-1 RA.' Discriminator — DPP-4i are weight-neutral and CV/renal-NEUTRAL, unlike GLP-1 RAs which produce weight loss and MACE reduction. If you need those outcomes, choose the injectable GLP-1 RA, not a gliptin.

7. Sulfonylureas & glinides

WHAT YOU SEE

Box 5: close K-ATP channel → Ca influx → insulin release; glinides before each meal; repaglinide.

WHAT IT ENCODES

Sulfonylureas (glipizide, glimepiride, glyburide) and meglitinides/glinides (repaglinide, nateglinide) close the beta-cell K-ATP channel → membrane depolarization → calcium influx → insulin exocytosis, INDEPENDENT of glucose level. They require functioning beta cells. SUs are long-acting (once-daily); glinides are short-acting, dosed BEFORE EACH MEAL to target postprandial glucose (useful with irregular meals). Mnemonic: 'shut the K channel, squeeze out insulin — meal-time for glinides.'

BOARD TRAP

Wrong answer: 'use a sulfonylurea or glinide in a type 1 diabetic / insulin-deficient patient.' Discriminator — secretagogues need residual beta-cell function, so they don't work in T1DM (no beta cells); they also cause hypoglycemia and weight gain. Use insulin in T1DM, and pick low-hypo agents when hypoglycemia is a concern.

8. Thiazolidinediones (TZD)

WHAT YOU SEE

Box 6: PPAR-gamma activation; increased insulin sensitivity; contraindicated in CHF; useful in PCOS.

WHAT IT ENCODES

Thiazolidinediones (pioglitazone, rosiglitazone) are PPAR-γ agonists that enhance peripheral insulin SENSITIVITY (adipose/muscle), lowering A1c without hypoglycemia and without renal dose adjustment. Useful where insulin resistance dominates (and historically in PCOS/NASH). Onset is slow (weeks). Mnemonic: 'PPAR-γ = sensitizer.'

BOARD TRAP

Wrong answer: 'pioglitazone is a safe add-on for a patient with NYHA III heart failure.' Discriminator — TZDs cause fluid retention/edema and are contraindicated in symptomatic HF; they also cause weight gain, fractures, and (rosiglitazone) CV concern/(pioglitazone) bladder cancer concern. Avoid in HF.

9. Acarbose

WHAT YOU SEE

Acarbose: brush-border alpha-glucosidase inhibitor; delays carbohydrate digestion; targets postprandial glucose.

WHAT IT ENCODES

Acarbose (and miglitol) inhibit intestinal brush-border alpha-glucosidase, delaying breakdown of complex carbohydrates/disaccharides into absorbable monosaccharides → blunts the POSTPRANDIAL glucose spike specifically. No hypoglycemia as monotherapy; modest A1c effect. Mnemonic: 'blocks carb digestion at the gut wall → flattens the after-meal peak.'

BOARD TRAP

Wrong answer: 'treat hypoglycemia in a patient on acarbose with orange juice or table sugar (sucrose).' Discriminator — acarbose blocks disaccharide breakdown, so sucrose won't be absorbed; you MUST give pure GLUCOSE (dextrose) to correct hypoglycemia. This is the signature acarbose board point.

10. Hypoglycemia on acarbose

WHAT YOU SEE

Hypoglycemia zone: dextrose works, sucrose does NOT.

WHAT IT ENCODES

Because acarbose prevents cleavage of sucrose into glucose + fructose at the brush border, oral sucrose (table sugar) is ineffective for hypoglycemia in a patient on acarbose — only monosaccharide GLUCOSE (dextrose tablets/gel, or IV dextrose) reliably raises blood glucose. Mnemonic: 'acarbose hypo → give glucose, not the sugar bowl.'

BOARD TRAP

Wrong answer: 'sucrose-containing candy is adequate rescue for acarbose hypoglycemia.' Discriminator — sucrose requires alpha-glucosidase to be absorbed, which acarbose blocks; reach for pure glucose/dextrose instead. (Note acarbose alone rarely causes hypoglycemia — usually when combined with insulin/secretagogues.)

11. Weight gain vs weight loss zones

WHAT YOU SEE

Weight gain: SU, glinide, insulin, TZD. Weight loss: GLP-1, SGLT2, metformin (neutral).

WHAT IT ENCODES

Weight effect by class: GAIN — sulfonylureas, glinides, thiazolidinediones, and insulin. LOSS — GLP-1 RAs and tirzepatide (most), SGLT2i (modest); metformin and DPP-4i are weight-NEUTRAL. This drives drug choice in the obese phenotype. Mnemonic: 'secretagogues + insulin + TZD add pounds; incretins + flozins shed them.'

BOARD TRAP

Wrong answer: 'add insulin or a sulfonylurea to an obese T2DM patient because they're potent.' Discriminator — both cause weight gain, worsening the obese phenotype; choose a GLP-1 RA/tirzepatide (or SGLT2i) for weight loss. Reserve insulin for severe hyperglycemia/insulinopenia.

12. HbA1c reduction ranking

WHAT YOU SEE

Front counter: metformin most potent among orals (tied with SU); A1c reduction ranking.

WHAT IT ENCODES

A1c-lowering potency: insulin and GLP-1 RAs (especially semaglutide/tirzepatide) are most potent overall; among orals, metformin and sulfonylureas give the largest reductions (~1–2%), TZDs ~1%, SGLT2i ~0.5–1%, and DPP-4 inhibitors the LEAST (~0.5–0.8%). Mnemonic: 'metformin/SU strong, gliptins weak.'

BOARD TRAP

Wrong answer: 'a DPP-4 inhibitor is the most powerful oral agent for lowering A1c.' Discriminator — DPP-4 inhibitors are the WEAKEST oral A1c-lowerers; metformin and sulfonylureas lower A1c most among orals. Don't pick a gliptin when maximal glucose lowering is the goal.